

Multiple Regression Model

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Ordinary Least Squares Estimation

Specification of the Model

Definition of OLS Estimator

Statistical Inference

The Population Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i, \quad i = 1, \dots, n$$

We have n observations, $(X_{1i}, X_{2i}, \dots, X_{ki}, Y_i), i = 1, \dots, n$

- Y_i is the dependent variable
- $(X_{1i}, X_{2i}, \dots, X_{ki})$ are the k independent variables or explanatory variables or regressors
- u_i is the error term
- β_0 is the intercept
- $\beta_1, \beta_2, \dots, \beta_k$ are the slope coefficients (or parameters)

The error term u_i captures all factors that could explain Y_i
over and above the explanatory variables $X_{1i}, X_{2i}, \dots, X_{ki}$

Interpretation of the slope coefficients $\beta_1, \beta_2, \dots, \beta_k$ is a bit different here (compared to the model with only one regressor)

For example, β_1 is the effect on Y of a unit change in X_1 , *holding* $X_2, X_3, \dots, X_k$ *constant*

Illustrating this via the PRF:

$$\begin{aligned} E[Y_i | X_{1i} = x_1, X_{2i} = x_2, \dots, X_{ki} = x_k] = \\ \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \end{aligned}$$

$$\begin{aligned} E[Y_i | X_{1i} = x_1 + 1, X_{2i} = x_2, \dots, X_{ki} = x_k] = \\ \beta_0 + \beta_1(x_1 + 1) + \beta_2 x_2 + \dots + \beta_k x_k, \end{aligned}$$

and therefore, the difference is equal to,

$$\begin{aligned} E[Y_i | X_{1i} = x_1 + 1, X_{2i} = x_2, \dots, X_{ki} = x_k] \\ - E[Y_i | X_{1i} = x_1, X_{2i} = x_2, \dots, X_{ki} = x_k] = \beta_1 \end{aligned}$$

The coefficient β_1 captures the so-called *partial effect on Y of X_1*
(And similarly, of course, for the other coefficients and regressors)

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Definition

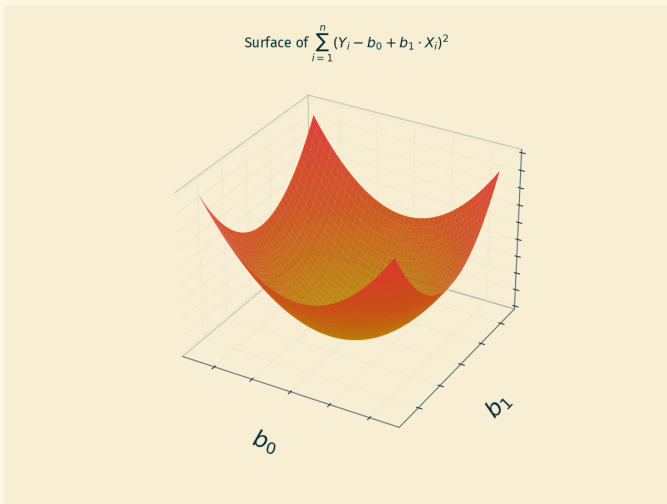
The **Ordinary Least Squares (OLS) estimators** are defined by

$$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k := \underset{b_0, b_1, \dots, b_k}{\operatorname{argmin}} \sum_{i=1}^n (Y_i - b_0 - b_1 X_{1i} - \dots - b_k X_{ki})^2$$

In words

- we look at the rhs as a function in $b_0, b_1, \dots, b_k$
- that function happens to be quadratic
- we find the values of $b_0, b_1, \dots, b_k$ that minimize that function
- the values that minimize that function are called solution
- we give the solution a specific name: $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$

Reminder: geometry of the minimization problem



But now the problem is higher dimensional!

In the multiple regression model, the geometry of the minimization problem cannot be illustrated in two dimensions because the problem is of dimension $k + 2$

(k slope coefficients plus constant term plus image dimension)

Going back to the mathematical problem:

Are you able to derive the solutions $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$?

Don't even bother!

Way too much work for too little extra insight

The best way to derive them by hand is by using matrix algebra

We rely on Python to compute the values for us

Predicted values and residual

Definition

The **predicted value** of Y_i is given by $\hat{Y}_i := \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \cdots + \hat{\beta}_k X_{ki}$.
The predicted value is the estimated PRF.

Definition

The **residual** is given by $\hat{u}_i := Y_i - \hat{Y}_i$.
It is the difference between Y_i and the predicted value.

These definitions are straightforward extensions of what you have learned for the simple regression model

Ordinary Least Squares Estimation

Statistical Inference

The Four Assumptions

Sampling Distribution of the OLS Estimators

Tests of Joint Hypotheses

Need to revisit the four OLS Assumptions for the multiple regression model

The assumptions are not all that different, they are mainly straightforward modifications of the ones we have seen for the simple regression model

OLS estimators are statistics that vary with the underlying random sample

Three natural questions to ask are:

- What is the expected value of the OLS estimator?
- What is the variance of the OLS estimator?
- What is the sampling distribution of the OLS estimator?

To answer these questions, we need to impose four assumptions

They are known as the OLS Assumptions

Assumption (OLS Assumption 1)

The error term u_i is conditionally mean independent (CMI) of $X_{1i}, \dots, X_{ki}$, meaning:

$$E[u_i | X_{1i}, \dots, X_{ki}] = E[u_i] = \mu_u.$$

Note: many textbooks simply set $\mu_u = 0$, which is without loss of generality

CMI restricts the expected value of the error terms

Although we do not observe the error terms u_i , and therefore we do not know their distribution, we are willing to impose a restriction on their expected values

The essential restriction in Assumption 1 is that the expected value of u_i *is not a function of* $X_{1i}, \dots, X_{ki}$

When we write that $E[u_i|X_{1i}, \dots, X_{ki}] = E[u_i]$, we are saying that $E[u_i|X_{1i}, \dots, X_{ki}]$

- is not dependent on $X_{1i}, \dots, X_{ki}$, and instead
- is constant with value μ_u

Assumption 1 says none of the $X_{1i}, \dots, X_{ki}$ is not informative for the mean of u_i

Assumption (OLS Assumption 2)

The sample data $(X_{1i}, \dots, X_{ki}, Y_i)$, $i = 1, \dots, n$, are independent and identically distributed (i.i.d.) draws from the joint population distribution.

Assumption 2 arises automatically if the entity i is sampled by simple random sampling:

- The entities are selected from the same population, so $(X_{1i}, \dots, X_{ki}, Y_i)$ are identically distributed for all $i = 1, \dots, n$
- The entities are selected at random, so the values of $(X_{1i}, \dots, X_{ki}, Y_i)$ for different entities are independently distributed.

The main place we will encounter non-i.i.d. sampling is when data are recorded over time for the same entity (panel data and time series data)

Assumption (OLS Assumption 3)

Large outliers in $X_{1i}, \dots, X_{ki}$ and/or Y_i are rare. Technically, $X_{1i}, \dots, X_{ki}$ and Y_i have finite fourth moments:

$$\text{for all } j = 1, \dots, k, \quad E[X_{ji}^4] < \infty, \quad E[Y_i^4] < \infty.$$

Large outliers for any of $X_{1i}, \dots, X_{ki}$ and Y_i can strongly influence OLS estimates

In practice, outliers are often data glitches
(coding or recording problems)

Simple suggestions:

- literally look at your data spreadsheet
- are there any suspicious numbers?
(for example, somebody's age was accidentally recorded as a negative number)
- do a scatterplot to spot outliers

Assumption (OLS Assumption 4b)

The error term u_i is **heteroskedastic**:

$$\text{Var}(u_i | X_{1i}, \dots, X_{ki}) = \sigma_u^2(X_{1i}, \dots, X_{ki})$$

We therefore explicitly allow the variance of the error term to be a function in the regressors

We always want to think of the error terms as heteroskedastic (rather than homoskedastic)

Reason: heteroskedasticity is much more general and Python can easily handle it for us easily

The textbook lists a completely different Assumption 4:

Assumption

*The regressors are not perfectly collinear.
(Or, there is no perfect multicollinearity.)*

I'm not a big fan of this assumption

Perfect collinearity between two regressors X_{hi} and X_{ji} means that they have a correlation of 1

But this would mean that $X_{hi} = X_{ji}$

But why would you ever include one and the same regressor twice on the rhs of your model?

This is not a problem at all in practice, so we just ignore this distraction

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The four OLS Assumptions are needed, as before, to derive the sampling distribution (or asymptotic distribution or large sample distribution) of $\hat{\beta}_1, \dots, \hat{\beta}_k$

For the simple regression model it was not a problem to do this by hand (although it was a bit tedious)

Here, in the multiple regression model, deriving the asymptotic distribution becomes impossible
(unless we resort to matrix algebra, which we won't)

Therefore, I will just state the results

Theorem

Under Assumptions 1 and 2 the OLS estimators $\hat{\beta}_1, \dots, \hat{\beta}_k$ are unbiased:

$$E[\hat{\beta}_j | X_{1i}, \dots, X_{ki}] = \beta_j, \quad \text{for } j = 1, \dots, k$$

Theorem

Under Assumptions 1, 2 and 4b, the OLS estimators have variances

$$\text{Var}(\hat{\beta}_j | X_{1i}, \dots, X_{ki}) = \sigma_{\hat{\beta}_j}^2, \quad \text{for } j = 1, \dots, k,$$

and covariances

$$\text{Cov}(\hat{\beta}_h, \hat{\beta}_j | X_{1i}, \dots, X_{ki}) = \sigma_{\hat{\beta}_h \hat{\beta}_j}$$

for $h = 1, \dots, k$, and $j = 1, \dots, k$.

In this theorem, the variances $\sigma_{\hat{\beta}_j}^2$ and the covariances $\sigma_{\hat{\beta}_h \hat{\beta}_j}$ are placeholders for complicated mathematical expressions

Theorem

The **asymptotic distribution of the OLS estimator** $\hat{\beta}_j$ under OLS Assumptions 1, 2, 3 and 4b is

$$\hat{\beta}_j \overset{\text{approx.}}{\sim} N\left(\beta_j, \sigma_{\hat{\beta}_j}^2\right),$$

for $j = 1, \dots, k$.

This theorem is the basis for deriving standard errors and confidence intervals for $\hat{\beta}_j$

But since we do not have an explicit result for what $\sigma_{\hat{\beta}_j}$ is and where it comes from, we will not provide explicit results for standard errors and confidence intervals

Instead, we will rely on Python to do the hard work for us

Example: association between test score and student teacher ratio in Californian secondary schools

So far, we've only done a simple regression of

- `testscr`: average student test score in a school district
- `str`: average number of students per teacher in a school district

Now we add two additional regressors:

- `el_pct`: the percentage of English learners in a school district (many students are migrants from Latin America)
- `expn_stu`: the total annual expenditure per student in the school district

Comparing simple versus multiple regression

Python Code (output edited)

```
> import pandas as pd
> import statsmodels.formula.api as smf
> df = pd.read_csv('caschool.csv')
> formula_simpreg = 'testscr ~ str'
> reg0 = smf.ols(formula_simpreg, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> print(reg0.summary())
```

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept    698.9330     10.364     67.436     0.000     678.619     719.247
str          -2.2798      0.519    -4.389     0.000     -3.298     -1.262
=====
```

Notes: [1] Standard Errors are heteroscedasticity robust (HC1)

```
> formula_multreg = 'testscr ~ str + el_pct + expn_stu'
> reg1 = smf.ols(formula_multreg, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> print(reg1.summary())
```

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept    649.5779     15.458     42.021     0.000     619.280     679.876
str          -0.2864      0.482     -0.594     0.552     -1.231      0.658
el_pct       -0.6560      0.032    -20.640     0.000     -0.718     -0.594
expn_stu       0.0039      0.002      2.447     0.014      0.001      0.007
=====
```

Notes: [1] Standard Errors are heteroscedasticity robust (HC1)

Modified interpretation of point estimate in multiple regression

Python Code (output edited)

```
> formula_multreg = 'testscr ~ str + el_pct + expn_stu'
> reg1 = smf.ols(formula_multreg, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> print(reg1.summary())
```

OLS Regression Results

	coef	std err	z	P> z	[0.025	0.975]
Intercept	649.5779	15.458	42.021	0.000	619.280	679.876
str	-0.2864	0.482	-0.594	0.552	-1.231	0.658
el_pct	-0.6560	0.032	-20.640	0.000	-0.718	-0.594
expn_stu	0.0039	0.002	2.447	0.014	0.001	0.007

Notes: [1] Standard Errors are heteroscedasticity robust (HC1)

- If `str` increases by one unit then `testscr` is expected to decrease by 0.2864 units holding everything else constant
- contrast to simple regression model:
If `str` increases by one unit then `testscr` is expected to decrease by 2.2798 units

Comparing the point estimates for `str` :

- simple regression: $\hat{\beta}_1 = -2.2798$, statistically significant
- multiple regression: $\hat{\beta}_1 = -0.2864$, statistically insignificant

If we view the multiple regression model as a 'richer' specification, then we could conclude that the simple regression model overstates (in absolute value) the association between test scores and the student teacher ratio

The simple regression model seems to ascribe too important a role to the student teacher ratio

When we control for other factors (adding `el_pct` and `expn_stu`), the importance of the student teacher ratio goes down

Aside: The highlighted numbers on the previous slide correspond to the standard errors $\hat{\sigma}_{\hat{\beta}_j}$

They are the estimates of the square root of the asymptotic variance $\sigma_{\hat{\beta}_j}^2$ from the previous theorem

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Recall the preceding multiple regression results:

Python Code (output edited)

```
> formula = 'testscr ~ str + el_pct + expn_stu'
> reg1 = smf.ols(formula, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> print(reg1.summary())
```

OLS Regression Results

	coef	std err	z	P> z	[0.025	0.975]
Intercept	649.5779	15.458	42.021	0.000	619.280	679.876
str	-0.2864	0.482	-0.594	0.552	-1.231	0.658
el_pct	-0.6560	0.032	-20.640	0.000	-0.718	-0.594
expn_stu	0.0039	0.002	2.447	0.014	0.001	0.007

You know how to use this output to test *simple* hypothesis

For example, if you want to test the hypotheses that the true coefficient of *expn_stu* is equal to zero, you would be testing $H_0 : \beta_3 = 0$ against the alternative $H_1 : \beta_3 \neq 0$

The t -statistic of $\hat{\beta}_3$ is 2.45 which is sufficiently larger than 1.96 and therefore we reject H_0 in favor of H_1

But what if you want to test *joint* hypotheses instead?

Let's say you want to test whether the two coefficients of *str* and *expn_stu* are both equal to zero

Formally, we want to test null hypothesis

$$H_0 : \beta_1 = 0 \qquad \text{and} \qquad \beta_3 = 0$$

$$H_1 : \beta_1 \neq 0 \qquad \text{and/or} \qquad \beta_3 \neq 0$$

You might think that it is ok to just look at their *t*-statistics separately

That is, you noticed that the t -statistic of $expn_stu$ is larger than 1.96 while the t -statistic of str is less than 1.96 (in absolute value) and therefore conclude that the joint hypothesis $\beta_1 = 0$ and $\beta_3 = 0$ does not appear to hold

Perhaps counter-intuitive, that kind of argument is flawed

The argument ignores that $\hat{\beta}_1$ and $\hat{\beta}_3$ and their t -statistics are likely to be correlated with each other

Testing β_1 and β_3 separately implicitly assumes that we can treat their estimators as statistically independent (they are not)

Instead of separately considering the t -statistic, one should look at the so-called F -statistic which implicitly recognizes the correlation between the t -statistics

In the given example, the F -statistic is given by

$$F := \frac{1}{2} \frac{t_1^2 + t_3^2 - 2\hat{\rho}_{t_1, t_3} t_1 t_3}{1 - \hat{\rho}_{t_1, t_2}^2},$$

where $\hat{\rho}_{t_1, t_3}$ is estimator of correlation b/w the two t -statistics

The F -statistic is a function of the two t -statistics

To get some intuition, set $\hat{\rho}_{t_1, t_3} = 0$

It follows that $F = (t_1^2 + t_3^2)/2$

This is the average of the two squared t -statistics

Under the null hypothesis $\beta_1 = 0$ and $\beta_3 = 0$, F is close to zero

Under the alternative hypothesis F will be a large number

Most generally, however, $\hat{\rho}_{t_1, t_3} \neq 0$ and the given formula for F also factors in that correlation between t_1 and t_3

Here is how you would test the joint hypothesis

$H_0 : \beta_1 = 0$ and $\beta_3 = 0$ in Python:

Python Code

```
> formula = 'testscr ~ str + el_pct + expn_stu'
> reg1 = smf.ols(formula, data=df, missing='drop').fit(cov_type='HC1', use_t=False)
> ftest = reg1.f_test('str = expn_stu = 0')
> print(ftest)

<F test: F=5.433727045036685, p=0.004682304362845301, df_denom=416, df_num=2>
```

The p-value equals 0.0047 which is smaller than 0.05 and we therefore reject H_0

Of course, you can also test any combination of coefficients in your model (not just β_1 and β_3)

For example, you may want to test *all* your coefficients

In the given example, we may want to test the joint hypothesis that

$$H_0 : \beta_1 = 0 \quad \beta_2 = 0 \quad \beta_3 = 0$$

against the alternative hypothesis that

$$H_1 : \text{at least one } \beta_j \neq 0 \text{ for } j = 1, 2, 3$$

To conduct this test, in Python you could simply type

Python Code

```
> ftest = reg1.f_test('str = el_pct = expn_stu = 0')  
> print(ftest)
```

```
<F test: F=147.20371132008665, p=5.201469651042804e-65, df_denom=416, df_num=3>
```

As you can see, the associated p-value is tiny

Therefore, we reject the null